

EPINAL-DOGNEVILLE AP, France

WMO: 071730

Lat: 48.2107N

Lon: 6.4517E

Elev: 1050

StdP: 14.15

Time Zone: 1.00 (EUC)

Period: 90-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	18.2	10.5	9.8	16.0	14.2	11.7	20.6	25.1	48.3	22.7	46.9	1.5	110	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	86.7	67.2	83.5	66.6	80.4	65.7	69.7	82.5	68.3	79.9	66.9	77.5	7.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.6	98.3	74.2	64.2	93.6	72.6	62.9	89.3	71.0	34.3	82.5	33.1	79.9	32.0	77.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	19.7	16.8	14.7	7.5	91.4	5.3	3.5	3.7	93.9	0.6	96.0	-2.4	98.0	-6.2	100.5	
(5)				7.9	72.8	5.8	1.0	3.7	73.5	0.3	74.1	-2.9	74.6	-7.1	75.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	34.5	36.7	42.5	48.4	55.6	62.4	65.5	64.9	58.6	51.5	41.8	34.9	(6)
(7)		DBStd	13.05	8.41	8.14	7.63	6.14	6.18	6.61	5.34	5.30	5.72	6.78	6.67	7.44	(7)
(8)		HDD50	2023	481	374	250	102	20	1	0	0	4	67	256	468	(8)
(9)		HDD65	5742	944	793	698	499	298	128	60	68	206	419	697	932	(9)
(10)		CDD50	1964	1	1	17	52	192	374	481	461	261	114	9	1	(10)
(11)		CDD65	210	0	0	0	0	6	51	77	64	12	0	0	0	(11)
(12)		CDH74	2587	0	0	0	16	101	571	936	814	146	3	0	0	(12)
(13)		CDH80	730	0	0	0	0	14	153	274	276	13	0	0	0	(13)
(14)	Wind	WSAvg	5.9	6.8	6.7	6.6	5.7	6.0	5.4	5.6	5.0	4.6	5.4	5.9	6.6	(14)
(15)	Precipitation	PrecAvg	44.5	4.0	3.5	3.2	3.0	3.8	3.4	3.4	3.5	3.5	4.2	4.4	4.6	(15)
(16)		PrecMax	59.5	9.1	6.7	9.3	7.5	6.3	6.5	7.4	7.4	6.5	8.2	10.8	10.5	(16)
(17)		PrecMin	31.2	1.0	0.8	0.4	0.6	0.8	1.1	1.0	0.4	0.9	0.7	0.9	0.6	(17)
(18)		PrecStd	6.4	2.0	1.8	2.2	1.7	1.3	1.4	1.5	1.7	1.8	2.0	2.4	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.3	60.1	67.2	76.3	82.0	89.2	90.8	95.9	82.1	73.6	62.0	52.4	(19)
(20)			MCWB	50.1	48.4	53.1	58.9	66.0	68.2	68.1	67.7	65.0	62.1	54.5	49.3	(20)
(21)		2%	DB	51.6	55.3	63.5	71.2	76.6	84.2	86.3	87.2	78.2	69.5	57.8	50.2	(21)
(22)			MCWB	48.4	47.1	51.5	56.7	61.0	67.2	67.2	66.6	64.7	60.2	52.7	48.3	(22)
(23)		5%	DB	49.0	52.2	59.2	66.7	73.3	80.7	83.2	82.6	74.6	66.0	54.7	48.1	(23)
(24)			MCWB	45.9	46.1	49.5	53.8	59.3	66.9	66.7	65.9	62.8	59.0	51.2	46.4	(24)
(25)		10%	DB	46.7	49.3	55.4	62.6	69.6	76.7	79.9	78.6	70.7	62.6	51.8	45.7	(25)
(26)			MCWB	43.9	45.2	48.5	52.2	58.0	65.3	65.7	64.9	61.4	57.6	49.2	44.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	50.2	55.5	61.4	67.9	71.3	71.2	70.9	68.6	63.8	56.2	50.2	(27)
(28)			MCDB	53.1	55.7	64.2	73.7	77.4	80.0	84.0	86.4	78.3	70.8	59.2	51.5	(28)
(29)		2%	WB	48.9	48.5	52.9	57.5	64.7	69.2	69.7	68.9	66.5	61.3	53.4	48.6	(29)
(30)			MCDB	51.2	53.1	59.7	68.6	74.4	80.3	83.4	81.7	75.1	67.1	57.0	50.2	(30)
(31)		5%	WB	46.2	47.0	51.1	55.1	61.2	67.6	68.1	67.3	64.5	59.6	51.4	46.6	(31)
(32)			MCDB	48.5	51.4	57.4	64.2	68.8	78.4	80.0	78.9	72.0	64.7	54.6	47.9	(32)
(33)		10%	WB	44.2	45.1	49.1	52.8	59.2	66.1	66.4	65.6	62.5	57.7	49.4	44.2	(33)
(34)			MCDB	46.5	49.0	54.9	60.7	66.6	75.8	76.9	75.7	68.9	61.9	51.7	45.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.9	13.9	17.4	20.5	20.0	20.0	20.2	20.4	19.9	16.4	11.5	9.4	(35)
(36)			MCDBR	9.2	21.4	26.2	30.7	27.9	27.9	28.5	31.0	26.7	22.0	16.5	8.8	(36)
(37)			MCWBR	7.1	14.0	16.0	17.1	14.1	12.6	12.1	13.0	14.0	13.7	11.7	7.4	(37)
(38)		5% WB	MCDBR	8.3	16.3	21.9	27.2	23.2	24.3	24.6	24.8	22.2	19.3	14.1	7.4	(38)
(39)			MCWBR	7.2	11.5	13.8	15.5	12.9	11.7	11.2	11.9	12.7	12.7	10.5	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.332	0.359	0.391	0.400	0.406	0.402	0.398	0.373	0.357	0.326	0.306	(40)	
(41)		taud	2.445	2.389	2.332	2.261	2.278	2.294	2.300	2.329	2.375	2.410	2.446	2.437	(41)	
(42)		Ebn at Noon	242	261	271	272	274	272	271	267	263	247	233	227	(42)	
(43)		Edh at Noon	21	27	33	40	41	40	39	37	32	26	21	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	307	550	941	1366	1610	1864	1748	1518	1158	699	354	247	(44)	
(45)		RadStd	46	94	145	193	174	160	167	152	115	57	38	45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page